

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B.Tech

Branch: Electrical Engineering

Semester: V

Subject Code: BTEEC501, Power System Analysis

Marks: 60

Date: 07/08/2023

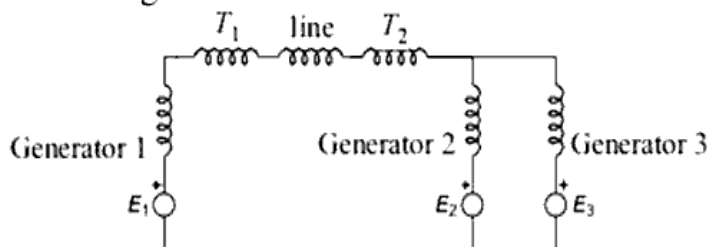
Duration: 3 Hours

Instructions to the students:

1. All questions are compulsory.
2. Figures to right indicate full marks.
3. Each question carries 12 marks.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

	CO/Level	Marks
Q.1) Attempt any two of the following questions.		12
a) Draw single-phase representation of balanced three-phase networks.	Lv.1	6
b) State Per Unit Representation of a Transformer.	Lv.1	6
c) Obtain the per unit impedance (reactance) diagram of the power system of Figure shown below.	Lv.2	6

Convert all reactance to per unit form. Choose a common three-phase MVA base of 30 and a voltage base of 33 kV line-to-line on the transmission line. Then the voltage base in the circuit of generator 1 is 11 kV line-to-line and that in the circuits of generators 2 and 3 is 6.2 kV.



Q.2) Attempt any two of the following questions.		12
a) Explain formation of Y_{bus} by graphical method by singular transformation.	Lv.3	6
b) State load flow problem and explain reference bus, load bus, generator bus and voltage control bus.	Lv.2	6
c) Comparison of Load Flow Methods.	Lv.1	6

Q.3)	Attempt any two of the following questions.		12
a)	Explain short circuit of a loaded synchronous machine.	Lv.3	6
b)	State selection criteria of circuit breakers.	Lv.1	6
c)	Derive sequence impedances of transmission lines.	Lv.2	6
Q.4)	Attempt any two of the following questions.		12
a)	Explain positive sequence impedances and networks of synchronous machine.	Lv.2	6
b)	State single line-to-ground (L-G) fault.	Lv.3	6
c)	Determine open conductor faults with two and one conductor open.	Lv.2	6
Q.5)	Attempt any two of the following questions.		12
a)	Explain security analysis in power system.	Lv.2	6
b)	Write note on,	Lv.1	6
	i. Small-disturbance Voltage Stability		
	ii. Voltage Stability		
	iii. Voltage Collapse		
c)	Explain methods to prevention of voltage collapse.	Lv.2	6

*** End ***